

## Updating methods improved the performance of a clinical prediction model in new patients

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### Abstract

**Objective:** Ideally, clinical prediction models are generalizable to other patient groups. Unfortunately, they perform regularly worse when validated in new patients and are then often redeveloped. While the original prediction model usually has been developed on a large data set, redevelopment then often occurs on the smaller validation set.

Recently, methods to update existing prediction models with the data of new patients have been proposed. We used an existing model that preoperatively predicts the risk of severe postoperative pain (SPP) to compare five updating methods.

**Study Design and Setting:** The model was tested and updated with a set of 752 new patients (274 [36] with SPP). We studied the discrimination (ability to distinguish between patients with and without SPP) and calibration (agreement between the predicted risks and observed frequencies of SPP) of the five updated models in 283 other patients (100 [35%] with SPP).

**Results:** Simple recalibration methods improved the calibration to a similar extent as revision methods that made more extensive adjustments to the original model. Discrimination could not be improved by any of the methods.

**Conclusion:** When the performance is poor in new patients, updating methods can be applied to adjust the model, rather than to develop a new model. © 2008 Elsevier Inc. All rights reserved.

**Keywords:** Prediction model; Testing; Updating; Recalibration; Calibration; Discrimination

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### 1. Introduction

Setting a diagnosis or a prognosis in clinical practice is a multivariable process. Physicians combine patient characteristics and test results to estimate the probability that a disease or outcome is present (diagnosis) or will occur (prognosis). To guide physicians in their estimation of diagnostic and prognostic probabilities, many clinical prediction models have been developed. These prediction models are algorithms that combine patient characteristics to set a diagnosis or to predict a prognosis. Such models can improve the identification of high-risk patients, and guide medical decision making. Besides that, they can be used as a quality of care instrument, because the predicted risks can be compared with the observed actual outcomes.

The performance of prediction models needs to be tested in new patients before the models can be applied in clinical

practice with confidence [1,2]. Unfortunately, the predictive performance is often decreased when a model is tested in new patients, compared to the performance estimated in the patients who were used to derive the model. As a consequence, the original prediction model is frequently rejected, and a new prediction model is developed. While the original prediction model usually has been developed on a large data set, redevelopment then often occurs on the much smaller data set of the new patients only. When every new patient sample leads to a new prediction model, the prior information that is captured in previous studies and prediction models is neglected. This is counterintuitive to the notion that research should be based on data of as many data patients as possible. The principle of using knowledge of previous studies has been recognized in etiologic and intervention research, in which cumulative meta-analyses are more common. The alternative to redeveloping prediction models in every new patient sample is updating the existing prediction models. The updated models combine the information that is captured in the original model with the information of the new patients [3–7]. As a result, the updated models are adjusted to the new patients.

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Recently, several updating methods have been proposed in the statistical literature [3]. The methods vary in extensiveness, which is reflected by the number of parameters that is adjusted or re-estimated. A relatively simple recalibration method, for instance, results in an updated model that has a new intercept and adjusted regression coefficients that are based on multiplication of the original coefficients with a single recalibration factor. With this method, only two parameters are estimated. More extensive updating methods estimate more parameters, for instance, when all individual regression coefficients are re-estimated.

In this study, we present an empirical example of updating a model to predict the risk of severe postoperative pain. We show the results of simple and more extensive updating methods.

## 2. Patients and methods

Severe postoperative pain occurs frequently after surgery. Patients at high risk of severe postoperative pain may benefit from preventative strategies. Therefore, a prediction model has been developed to predict the risk of acute severe postoperative pain.

### 2.1. The data sets

Usually, prediction models are developed in derivation sets and tested in validation sets. However, because we wanted to show the effect of the updating methods, the validation set was split into an updating set and a test set (in which the updated model is tested). As a result, three data sets were used in this study (Fig. 1).

#### 2.1.1. The derivation set

The data set that was used to derive the prediction model for severe postoperative pain in surgical patients included 1,944 surgical patients (1,395 inpatients and 549 outpatients) [8]. In brief, the patients, aged 18–85, were selected from a randomized trial to investigate whether intravenous

anesthesia (propofol) yielded a lower incidence of postoperative nausea and vomiting than inhalation anesthesia (isoflurane and nitrous oxide) [9]. The trial was conducted at the Academic Medical Center of the University of Amsterdam, The Netherlands between April 1997 and January 1999. All types of surgery were included, except cardiac surgery and intracranial neurosurgical procedures. Exclusion criteria were emergency surgery, pregnancy, ASA physical status 4, and morbid obesity (weight > 120 kg).

ASA physical status denotes the American Society of Anesthesiologists preoperative risk classification based on comorbidity, and ranges from 1 (healthy patient) to 4 (patient with comorbidity that is a constant threat to life). Besides postoperative nausea and vomiting, the severity of acute postoperative pain was recorded with a numerical rating scale (NRS) score (where 0 indicates no pain at all, and 10 the most severe pain imaginable) every 15 minutes after arrival at the Post Anaesthesia Care Unit (PACU) for one hour (four measurements). Severe postoperative pain was defined as an NRS score of 6 or higher on at least one of the four measurement points.

#### 2.1.2. The updating and test sets

The updating and test sets together contain 1,035 adult surgical patients (591 inpatients and 444 outpatients) scheduled for surgery in 2004 in a second hospital, the University Medical Center of Utrecht, the Netherlands (Fig. 1). Similar as in the derivation set, all types of surgery were included, except emergency surgery, cardiac surgery, and intracranial neurosurgical procedures. Also, the same methods of data collection were applied regarding the predictors. However, a small change was made to the assessment of the outcome for logistic reasons. The NRS of the inpatients was scored 15 and 60 minutes after arrival at the PACU. The NRS of the outpatients was scored at arrival and after 15, 30, and 45 minutes on the PACU.

A test set should include at least 100 events to contain reasonable power to detect changes in predictive performance of prediction models between two data sets [10].

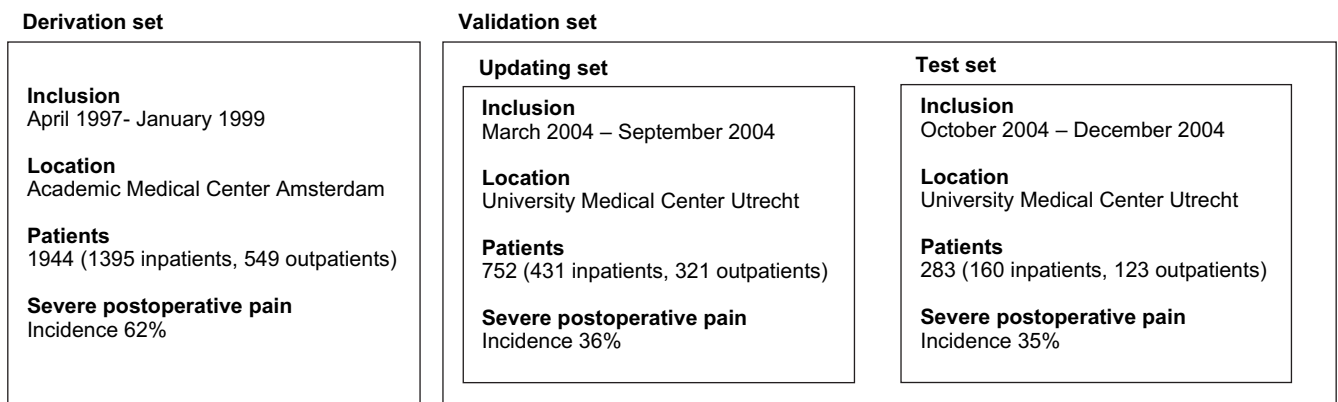


Fig. 1. Schematic presentation of the characteristics of the derivation, updating, and test data sets.

We therefore included 283 patients in the test set, of which 100 patients had severe postoperative pain, and 183 patients had no severe postoperative pain (outcome incidence 35%). To comply with the chronology in practice, we did not randomly select these 283 patients in the test set, but rather the more recently treated patients.

## 2.2. The prediction model

The prediction model predicts the risk of severe acute postoperative pain in the first hour after surgery.

The model was developed by multivariable logistic regression and included eight predictors that were assessed during the preoperative visit within a few weeks before surgery: gender, age, type of surgery, expected incision size, preoperative pain, anxiety score, need for information score, and type of patient. The classification of type of surgery (lowest pain, low pain, moderate pain, high pain, and highest pain) was developed in the derivation study, based on clinical experience of surgeons and anesthesiologists, and the associations we found in the data [8]. The expected incision size of the surgical procedure was dichotomized into smaller than 10 cm and equal to or larger than 10 cm. The severity of preoperative pain was assessed with the pain domain of the quality-of-life questionnaire Short Form 36 (SF-36), that was transformed to a score of 0 (no pain) to 10 [11]. The anxiety score and need for information score were assessed with the Amsterdam Preoperative Anxiety and Information Scale (APAIS). The APAIS consists of six questions, each scored on a 5-point Likert scale from 1 (not at all) to 5 (extremely). Four questions are used to assess the patients' preoperative anxiety score (range 4–20), and two questions are used to assess the patients' need for information regarding the scheduled surgery and anesthesia (range 2–10) [12]. Type of patient was defined as inpatient or outpatient. Interaction terms were included because the effect of some predictors was different for inpatients and outpatients [8].

Most predictors were completely observed for all patients, except incision size (one value missing, 0.07% of patients), preoperative pain (38 values missing, 2.7% of patients), anxiety (27 values missing, 1.9% of patients), and need for information (34 values missing, 2.4% of patients).

The formula of the prediction model is of the form:

$$\log\left(\frac{\text{risk of pain}}{1 - \text{risk of pain}}\right) = \text{linear predictor} \\ = \beta_0 + \beta_1 \times \text{predictor}_1 + \dots + \beta_n \times \text{predictor}_n \quad (1)$$

In this formula,  $\beta_0$  is the intercept and  $\beta_1$  till  $\beta_n$  are the regression coefficients, which are presented in Table 1.

The risk of severe postoperative pain in individual patients (scale: 0%–100%) can be calculated with the formula:  $\text{risk} = 1/(1 + e^{-\text{linear predictor}})$ . The model was also presented as an easy to use score chart [8]. The area under

Table 1

Intercept and regression coefficients of the prediction model for severe postoperative pain

| Predictor                                                    | Regression coefficient |
|--------------------------------------------------------------|------------------------|
| Intercept                                                    | −0.42                  |
| Female gender                                                | −0.004                 |
| Age                                                          | −0.009                 |
| Type of surgery                                              |                        |
| Lowest expected pain (reference)                             | —                      |
| Low expected pain                                            | 0.50                   |
| Moderate expected pain                                       | 0.92                   |
| High expected pain                                           | 1.05                   |
| Highest expected pain                                        | 1.72                   |
| Expected incision size $\geq 10$ cm                          | 0.39                   |
| Preoperative pain score                                      | 0.11                   |
| Anxiety score                                                | 0.05                   |
| Need for information score                                   | −0.05                  |
| Ambulatory surgery                                           | −0.70                  |
| Ambulatory surgery $\times$ female gender                    | 0.67                   |
| Ambulatory surgery $\times$ type of surgery                  |                        |
| Ambulatory surgery $\times$ lowest expected pain (reference) | —                      |
| Ambulatory surgery $\times$ low expected pain                | −0.10                  |
| Ambulatory surgery $\times$ moderate expected pain           | −0.47                  |
| Ambulatory surgery $\times$ high expected pain               | −0.07                  |
| Ambulatory surgery $\times$ highest expected pain            | −1.51                  |

the receiver-operating characteristic (ROC) curve (AUC) of the model was 0.71 in the derivation set.

## 2.3. Analyses

### 2.3.1. Missing values

Deleting subjects with missing values often leads to bias and a loss of statistical power [13–15].

All missing values were imputed with multiple imputation techniques using all other available information of the patients (aregImpute in the Design Library, standard available in S-plus software version 6.2). Logistic regression models were used to impute dichotomous missing values, polytomous logistic regression models for categorical missing values, and linear regression models for continuous missing values.

### 2.3.2. Predictive performance

To study the performance of the prediction model in the updating and test set, we assessed its calibration and discrimination. Calibration refers to the agreement between the predicted risks and observed frequencies of severe postoperative pain. This was graphically assessed with a calibration plot that shows a calibration line, which can be described with a slope and an intercept (calibration slope and calibration intercept) [16,17]. These parameters were first estimated in the updating set using a logistic regression model with the presence of severe postoperative pain (yes/no) as the outcome variable and the linear predictor of the original prediction model [Equation (1)] as the only

covariate. The calibration slope and calibration intercept are ideally 1 and 0, respectively (perfect calibration). Because a calibration intercept is difficult to interpret when the calibration slope is not equal to 1, the calibration intercept is estimated with the calibration slope fixed at 1. This is done by fitting the logistic regression model with the regression coefficient of the linear predictor fixed at 1. A calibration intercept different from 0 indicates that the model's predicted probabilities in the updating set are systematically too high (intercept < 0) or too low (intercept > 0). Such a difference in calibration intercept commonly reflects a difference in outcome incidence between the derivation set and the updating set that cannot be explained by different distributions of the predictor values. A calibration slope smaller than 1 indicates optimism; the regression coefficients of the original model were too large, which results in too extreme predictions in the new patients (low predictions are too low and high predictions are too high). A calibration slope that is larger than 1 indicates that the regression coefficients of the original model were too close to zero.

Discrimination is the ability of the model to discriminate patients with severe pain from patients without severe pain. It was quantified with the AUC, which is equal to the c-statistic for a dichotomous outcome variable [18].

### 2.3.3. Updating methods

Several methods can be used to update a prediction model when it shows poor performance in new patients. The methods we used are shown in Table 2 and have been presented previously in the statistical literature [3].

In method 0, no adjustments are made to the original prediction model [Equation (1)]. Method 1 and method 2 are so-called “recalibration” methods. Method 1 intends to correct the “calibration in the large,” such that the mean predicted probability is equal to the observed outcome frequency [3]. Only the intercept of the original model is adjusted [4,5]. This can be done by fitting a logistic regression model with the linear predictor (as offset) as the only covariate in the updating set or by calculating a correction factor [Equation (2)] that is based on the mean predicted risk and observed outcome frequency in the updating set.

$$\text{correction factor} = \ln \left( \frac{\frac{\text{observed outcome frequency}}{1 - \text{observed outcome frequency}}}{\frac{\text{mean predicted risk}}{1 - \text{mean predicted risk}}} \right) \quad (2)$$

This correction factor equals the calibration intercept when the outcome frequency is not extremely low or high. The correction factor simply needs to be added to the intercept of the original prediction model, which results in a new intercept.

In method 2, both the calibration intercept and the calibration slope are estimated with Equation (3) to adjust the original model, also called “logistic calibration” [13]. A

Table 2

Updating methods for the prediction rule for severe acute postoperative pain

| No. | Updating method                                                                                                                               |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | No adjustment (the original prediction rule)                                                                                                  |
| 1   | Adjustment of the intercept using the calibration intercept                                                                                   |
| 2   | Adjustment of the intercept and the regression coefficients using the calibration intercept and calibration slope                             |
| 3   | Extra adjustment of predictors with a different effect in the updating set compared to the derivation set, after recalibration by method 2    |
| 4   | Re-estimation of the intercept and the regression coefficients of all predictors, using the data of the updating set                          |
| 5   | Re-estimation of the intercept and the regression coefficients of all predictors, using the combined data of the derivation and updating sets |

Note: Methods 1 and 2 are recalibration methods, whereas methods 3, 4, and 5 are revision methods.

logistic regression model is fitted with the linear predictor as the only covariate in the updating set.

$$\ln \left( \frac{\text{risk of pain}}{1 - \text{risk of pain}} \right) = \alpha_{\text{calibration}} + \beta_{\text{calibration}} \times \text{linear predictor} \quad (3)$$

The calibration slope  $\beta_{\text{calibration}}$  is used to recalibrate the original regression coefficients, that is, the regression coefficients of the original model are multiplied with the calibration slope. A calibration slope equal to 1 means that the original regression coefficients do not need adjustment. The intercept of the original prediction model is adjusted by adding the calibration intercept  $\alpha_{\text{calibration}}$ .

Updating methods 3, 4, and 5 are so-called revision methods. Method 3 builds on method 2. It tests whether the effect of the individual predictors is different in the updating set compared to the derivation set, after the logistic calibration of method 2.

$$\ln \left( \frac{\text{risk of pain}}{1 - \text{risk of pain}} \right) = \alpha_{\text{calibration}} + \beta_{\text{calibration}} \times \text{linear predictor} + \gamma \times \text{predictor} \quad (4)$$

The logistic regression model of Equation (4) is first fitted for each predictor of the original model separately. The  $\gamma$  is the deviation from the recalibrated regression coefficient (based on method 2). When  $\gamma$  is not equal to 0, the effect of the predictor is still different in the updating set, after the recalibration of method 2. We test with the likelihood ratio test ( $P < 0.05$ ) whether the deviation has added predictive value. We selected the predictors with forward selection, starting with the predictor with the largest Wald statistic. We note that a “predictor” in Equation (4) can also be an interaction term.

In methods 4 and 5, the intercept and the regression coefficients of all predictors in the original model are re-estimated, without variable selection. In method 4, only



the updating set was used, and in method 5 the combined derivation and updating set. Because the incidence in the derivation set was quite different from that in the updating and test set (62%, 36%, and 35%, respectively), the incidence in the combined derivation and updating set was higher than that in the test set. Therefore, for method 5, the intercept of the updated model was first adjusted by applying updating method 1.

#### 2.3.4. Shrinkage methods

Newly estimated individual regression coefficients (such as in methods 3, 4, and 5) are often too optimistic, due to overfitting. This can be overcome by shrinkage of the regression coefficients. Regression coefficients can be shrunk either toward zero [3,19] or toward the recalibrated values (the regression coefficients of the model updated with method 2). Shrinkage toward zero needs to be applied when no information is present about the strength of the effect of the predictors in the model [19]. For method 5, where we combined the derivation set and the updating set and ignored the regression coefficients of the original model, all regression coefficients were in fact estimated without prior knowledge. As a result, the regression coefficients of the model updated with method 5 were shrunk toward zero using the following formula:

$$\beta_{\text{method5}} = \text{shrinkage factor} \times \beta_{\text{re-estimated}} \quad (5)$$

In this formula, the shrinkage factor is a heuristic shrinkage factor [19], calculated as:

$$\text{shrinkage factor} = \frac{\text{model } \chi^2_{\text{updated-nullmodel}} - df}{\text{model } \chi^2_{\text{updated-nullmodel}}} \quad (6)$$

in which  $\chi^2_{\text{updated-nullmodel}}$  is the difference in  $-2\log\text{likelihood}$  of a model with only an intercept and the updated model, and  $df$  equals the difference in degrees of freedom between these models.

Shrinkage toward the recalibrated values can be applied when there is prior knowledge about the regression coefficients. In method 3, the same predictors and regression coefficients were used as in the original model. The only difference was that the deviation of the predictors between the derivation and updating sets was modeled. Therefore, the deviations of the regression coefficients of the model updated with method 3 are shrunk toward the recalibrated values [3]. The regression coefficients of method 3 were shrunk with the formula:

$$\beta_{\text{method3}} = \beta_{\text{recalibrated}} + \text{shrinkage factor} \times \gamma \quad (7)$$

In this formula,  $\beta_{\text{recalibrated}}$  is the recalibrated regression coefficient (based on method 2), and  $\gamma$  is the deviation after updating with method 3 from the recalibrated coefficient. The shrinkage factor can be calculated as:

$$\text{shrinkage factor} = \frac{\text{model } \chi^2_{\text{updated-recalibrated}} - df}{\text{model } \chi^2_{\text{updated-recalibrated}}} \quad (8)$$

where  $\chi^2_{\text{updated-recalibrated}}$  is the difference in  $-2\log\text{likelihood}$  between a model with estimated deviations of individual regression coefficients and the recalibrated model, and  $df$  stands for the difference in degrees of freedom between these models.

Also for method 4, shrinkage toward the recalibrated values could be applied as there was prior knowledge about the regression coefficients. The same predictors were used as in the original model, only the regression coefficients were re-estimated. Hence, the regression coefficients of method 4 were shrunk with the formula:

$$\beta_{\text{method4}} = \beta_{\text{recalibrated}} + \text{shrinkage factor} \times (\beta_{\text{re-estimated}} - \beta_{\text{recalibrated}}) \quad (9)$$

In this formula,  $\beta_{\text{re-estimated}}$  is the re-estimated regression coefficient. The shrinkage factor can be calculated from Equation (8).

### 3. Results

Approximately one-third of the patients in the updating (274, 36%) and test sets (100, 35%) reported severe postoperative pain in the first hour after surgery, compared to 62% (1,205) of the patients in the derivation set (Table 3). Patients in the updating and test sets were slightly older, they underwent more often ambulatory surgery, and had less often a large expected incision size than patients in the derivation set. The distribution of patient characteristics in the updating and test sets was very similar, although the test set included more women (183, 65%) than the updating set (413, 55%), which must be due to chance.

#### 3.1. Predictive performance of the original prediction model (method 0)

The calibration plot showed that the predicted probabilities of the original prediction model were systematically too high (bias) in the updating set, as indicated by the solid line (Fig. 2). The calibration slope was 0.76, indicating that the originally estimated regression coefficients were too large, resulting in too extreme predictions in the new patients. The calibration intercept, given slope = 1, was  $-0.98$ , indicating that the predicted probabilities were systematically too high. The discriminative ability of the model was lower in the updating set ( $\text{AUC} = 0.65$  [0.57–0.73]), than in the derivation set (0.71 [0.66–0.76]). Given the poor calibration and lower AUC, an updated prediction model may have better predictive performance in new patients.

#### 3.2. The updated prediction models

Table 4 shows the estimates of the main parameters of the various updating methods. The regression coefficients of all predictions per updated model are presented in the

Table 3

Distribution of the predictors in the derivation set, the updating set, and the test set, *n* (%) unless stated otherwise

| Patient characteristics                  | Derivation set<br>( <i>n</i> = 1,944) | Updating set<br>( <i>n</i> = 752) | Test set<br>( <i>n</i> = 283) |
|------------------------------------------|---------------------------------------|-----------------------------------|-------------------------------|
| Female gender                            | 1,110 (57%)                           | 413 (55%)                         | 183 (65%)                     |
| Age (yr) <sup>a</sup>                    | 43 (15%)                              | 46 (16)                           | 48 (16)                       |
| Ambulatory surgery                       | 549 (28%)                             | 321 (43%)                         | 123 (43%)                     |
| Preoperative pain score <sup>a</sup>     | 3.0 (2.7)                             | 3.4 (2.9)                         | 3.3 (2.9)                     |
| Type of surgery                          |                                       |                                   |                               |
| Lowest pain                              | 72 (4%)                               | 61 (8%)                           | 31 (11%)                      |
| Low pain                                 | 590 (30%)                             | 216 (29%)                         | 81 (29%)                      |
| Moderate pain                            | 348 (18%)                             | 139 (18%)                         | 61 (22%)                      |
| High pain                                | 671 (35%)                             | 252 (34%)                         | 87 (31%)                      |
| Highest pain                             | 263 (14%)                             | 84 (11%)                          | 23 (8%)                       |
| Expected incision size ≥ 10 cm           | 724 (37%)                             | 50 (7%)                           | 17 (6%)                       |
| APAIS anxiety score <sup>a</sup>         | 9.4 (4.0)                             | 9.5 (3.8)                         | 9.7 (4.0)                     |
| APAIS need for information <sup>a</sup>  | 6.5 (2.2)                             | 6.2 (2.3)                         | 6.2 (2.4)                     |
| Severe acute postoperative pain, NRS ≥ 6 | 1205 (62%)                            | 274 (36%)                         | 100 (35%)                     |

<sup>a</sup> Mean (standard deviation).

**Appendix.** In method 1, only the model intercept was updated. Corresponding with the too high predicted risks in the updating set, the intercept of the updated model was decreased with  $-0.98$ , in accordance with the estimated calibration intercept described above. In method 2 (adjusting the intercept and regression coefficients by the calibration intercept and calibration slope, respectively), the individual regression coefficients were multiplied with  $0.76$ , and the intercept of the original prediction model was decreased with  $-0.77$ . This implicates that although the calibration

slope corrects the original regression coefficients (that were too large), an adjustment in the intercept was still needed. However, the adjustment to the intercept of the original model was smaller ( $-0.77$ ) than the adjustment that was needed without the adjustment of the regression coefficients with the calibration slope ( $-0.98$  in method 1). The updated regression coefficients are calculated by multiplying the calibration slope of method 2 ( $0.76$ ) with the regression coefficient of the original model. For ambulatory surgery, for instance, this results in a regression coefficient of  $0.76(-0.70) = -0.53$ .

Method 3 (revision of the predictors with a different association in the updating set than in the derivation set) showed a statistically significant different effect of the interaction between ambulatory surgery (outpatients versus inpatients) and type of surgery between the updating set and the derivation set ( $P < 0.001$ ). The effects of the other predictors were not significantly different. The updated regression coefficient of the interaction between ambulatory surgery and type of surgery, for instance, the interaction between low pain procedure and ambulatory surgery, was obtained by multiplying the calibration slope ( $0.90$ ) with the regression coefficient of the original model ( $-0.10$ ).

The deviation from this regression coefficient ( $-0.54$ , Table 4) was then added, after shrinkage with the shrinkage factor ( $0.89$ , Table 4). This resulted in  $0.90(-0.10) + 0.89(-0.54) = -0.57$  as the updated regression coefficient of the interaction between low pain procedure and ambulatory surgery (see Appendix). The regression coefficient of the interaction between low pain procedure and ambulatory surgery was more negative than that in the derivation set (deviation  $-0.47$ ). Outpatients who are scheduled for low pain procedures thus had a lower risk of severe postoperative pain in the updating set than outpatients in the derivation set. The deviation of the other types of surgery was in an opposite direction, which indicates that for outpatients who were scheduled

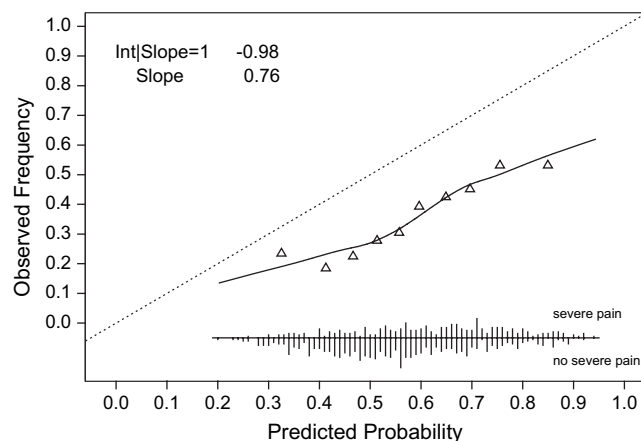


Fig. 2. Calibration plot of the original prediction model tested in the updating set (updating method 0).

Table 4

Estimated parameters of the updating methods 1, 2, and 3 (see text for more information about the updating methods) and the shrinkage factor for methods 3, 4, and 5

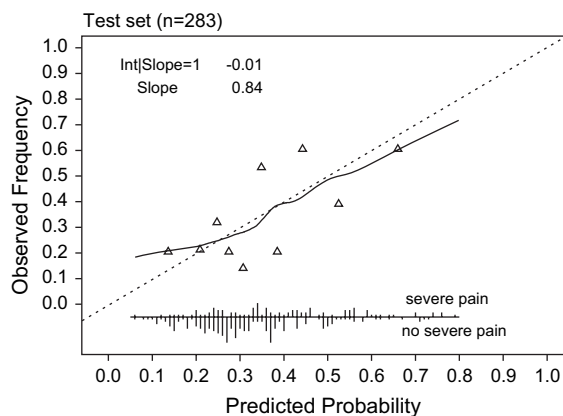
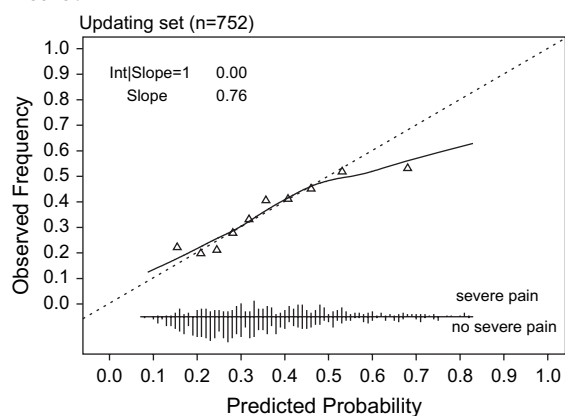
|                                                                                                                 | Method 1  | Method 2 | Method 3 | Method 4 | Method 5 |
|-----------------------------------------------------------------------------------------------------------------|-----------|----------|----------|----------|----------|
| Calibration intercept                                                                                           | $-0.98^a$ | $-0.77$  | $-1.01$  | —        | —        |
| Calibration slope                                                                                               | —         | $0.76$   | $0.90$   | —        | —        |
| Deviation from recalibrated regression coefficients between ambulatory surgery and type of surgery <sup>b</sup> |           |          |          |          |          |
| Low pain                                                                                                        | —         | —        | $-0.54$  | —        | —        |
| Moderate pain                                                                                                   | —         | —        | $0.99$   | —        | —        |
| High pain                                                                                                       | —         | —        | $0.35$   | —        | —        |
| Highest pain                                                                                                    | —         | —        | $2.65$   | —        | —        |
| Shrinkage factor                                                                                                | —         | —        | $0.89$   | $0.71$   | $0.96$   |

*Note:* The re-estimated parameters of methods 4 and 5 are not presented in this table. The (re-estimated) regression coefficients of all updated models are presented in the Appendix.

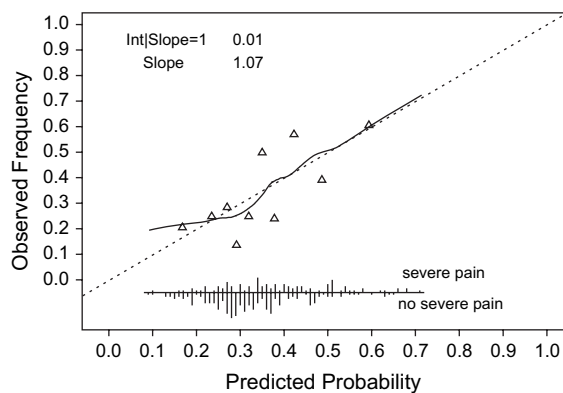
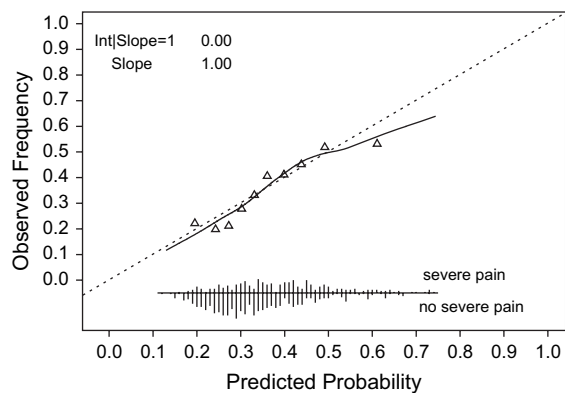
<sup>a</sup> Calibration intercept with calibration slope fixed at 1.

<sup>b</sup> Before shrinkage.

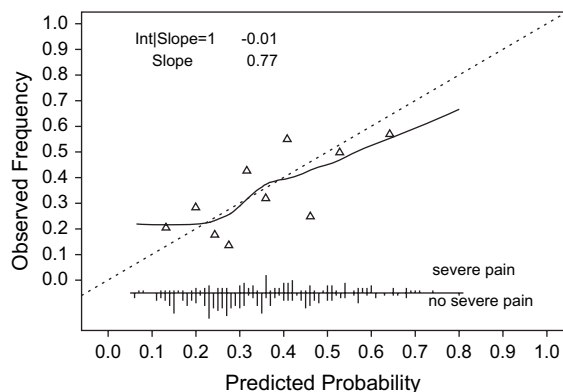
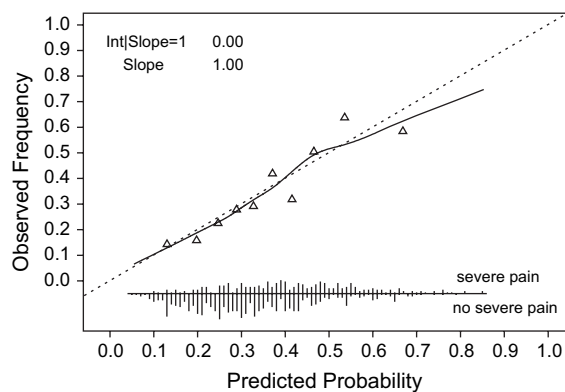
## Method 1



## Method 2



## Method 3



## Method 4

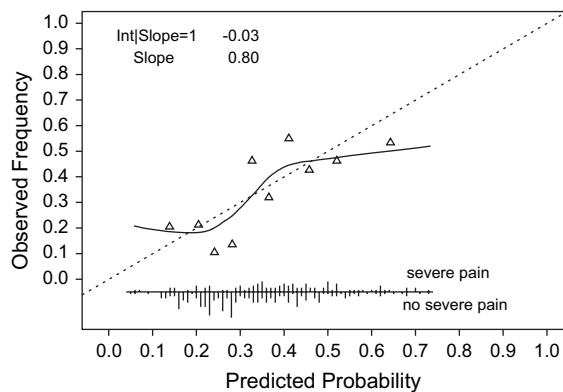
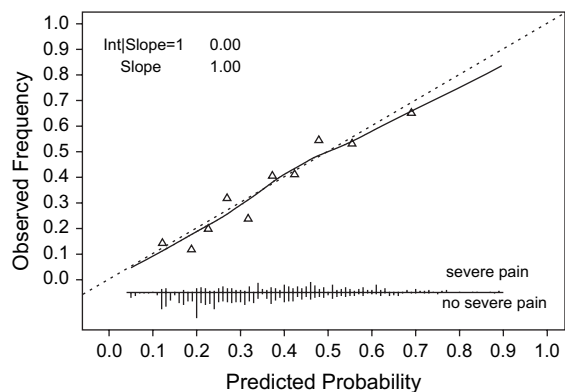


Fig. 3. Calibration plots of the prediction model that was updated by methods 1,2,3,4, and 5. Calibration plots are shown for the updated models in the updating set and the test set. The updating methods are described in the text. For method 5, the derivation and updating sets were combined. Because the incidence of severe postoperative pain in the combined set is higher than that in the test set, also the calibration plot of method 5 with an adjusted intercept is presented.

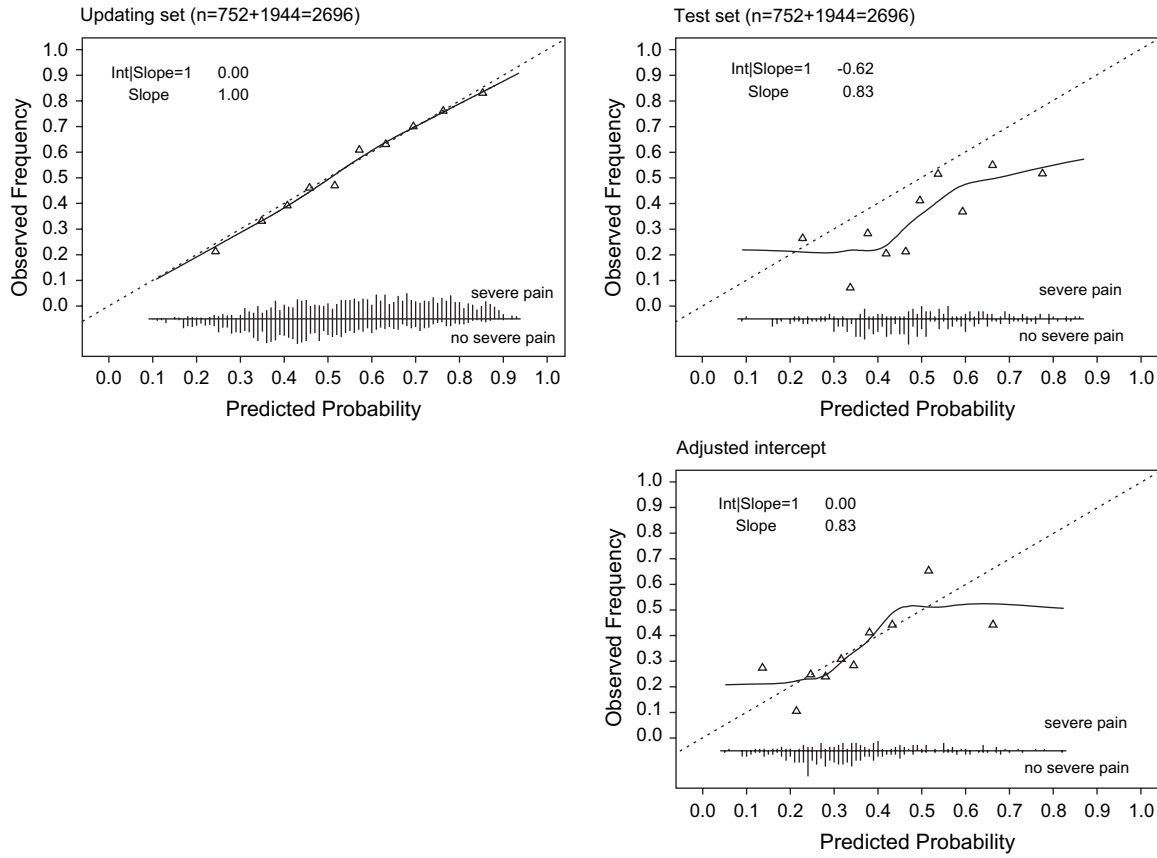
**Method 5**

Fig. 3. (continued)

for procedures with moderate, high, or highest pain, the risk of severe postoperative pain was higher for patients in the updating set than those in the derivation set.

### 3.3. Predictive performance of the updated prediction models in the updating and test sets

All updated models resulted in improved calibration in the updating set, compared to the original model (Fig. 3, left part). Because this set was also used to update the model, intercept and slope were always equal to 0 and 1, respectively, by definition (except for the slope of the model updated with method 1 as here only the model's intercept was updated). Methods 1 to 4 showed all good calibration in the large (mean predicted risk equals observed incidence) when the models were tested in the test set. This

is reflected by a calibration intercept close to zero (Fig. 3, right part). This also applies to method 5, though only after extra adjustment of the intercept. The calibration slopes showed more variability between the updated models in the test set. In method 1, only the calibration intercept was used to update the model, and the updated model showed a calibration slope of 0.84. Method 2 resulted in a calibration slope of 1.07 (close to 1), whereas adjustment of the individual regression coefficients with methods 3 to 5 did not result in better calibration slopes (0.77, 0.80, and 0.83, respectively).

Table 5 shows that the models updated with recalibration methods 1 and 2 obviously had the same AUC as the original prediction model in the updating set (0.65). These updating methods do not change the ranking of the predicted risks of patients, and thus do not affect a model's discriminative

Table 5

Discrimination of the updated prediction models in the updating set and test set, expressed by the AUC (95% confidence interval)

|                                         | Method 0 <sup>a</sup> | Method 1         | Method 2         | Method 3         | Method 4         | Method 5         |
|-----------------------------------------|-----------------------|------------------|------------------|------------------|------------------|------------------|
| Updating set ( $n = 752$ ) <sup>b</sup> | 0.65 (0.57–0.73)      | 0.65 (0.57–0.73) | 0.65 (0.57–0.73) | 0.70 (0.62–0.77) | 0.71 (0.63–0.79) | 0.72 (0.68–0.75) |
| Test set ( $n = 283$ )                  | 0.66 (0.52–0.80)      | 0.66 (0.52–0.80) | 0.66 (0.52–0.80) | 0.66 (0.52–0.79) | 0.67 (0.54–0.80) | 0.67 (0.53–0.80) |

Abbreviations: AUC, area under the receiver-operating curve; 95% CI, 95% Confidence interval.

<sup>a</sup> Original prediction rule.

<sup>b</sup> For method 5, the derivation and updating sets are combined into a data set containing 2,696 patients.



ability. Updating methods 3, 4, and 5 resulted in an increased AUC in the updating set (0.70, 0.71, and 0.72, respectively). However, when tested in the test set, this increase in AUC did not hold, and the updated models showed a similar AUC as the original prediction model in the test set (0.66).

#### 4. Discussion

We tested a prediction model for the risk of acute severe postoperative pain in surgical patients. The patients were recruited in a later period and in another hospital than the patients on which the model was developed. As the predictive performance of the model was poor, updating was necessary to adjust the model to the new situation. We compared five updating methods that differed in extensiveness of the updating. In this example study, simple recalibration methods improved the calibration to a similar extent as the revision methods that made more extensive adjustments to the original model. Discrimination was improved in the updating set when the more extensive updating methods were applied, but this improvement did not hold in the test set. We compared updating methods in one empirical data set. Other data sets can show different results. In general, when the discrimination of a model is sufficient, recalibration methods can improve the calibration. On the other hand, when the discrimination of a model needs to be improved, revision methods are necessary. However, the aim of our example study was not merely to present the best method, but above all to increase the familiarity and appreciation of updating methods of prediction models.

The predicted risks in new patients can be biased, that is, systematically over- or underestimated [3,20]. This can, for instance, be caused by variables or characteristics that are not included in the model, but do have an effect on the originally estimated regression parameters (intercept and regression coefficients). Although researchers in such situations often pursue to completely re-estimate all regression coefficients using the data of the new patients only (method 4), we showed that this did not improve the performance of the model more than the simple recalibration methods (methods 1 and 2) in our example study. Moreover, when researchers expect that a particular predictor in the original model indeed has a different effect in the updating set, revision methods with small adjustments for that predictor (method 3) are still preferred over methods that re-estimate all regression coefficients with only the new data (method 4). Another option when re-estimating the regression coefficients is to combine the derivation and updating sets, if one has indeed access to the derivation data (method 5). However, in our example study, this did also not result in a better performance in the test set than the simple recalibration methods (methods 1 and 2). Our results support the view that researchers should opt first for adjusting the existing model to the new patients by estimating

only a few parameters (methods 1 and 2, and perhaps 3) rather than directly re-estimating or rebuilding a complete new model (methods 4 and 5) [3,20].

Choosing between simple and more extensive updating methods is a choice between variance and bias, in which the size of both the derivation and the updating sets should be considered. Estimates of regression parameters of prediction models derived on large derivation sets are obviously very precise (low variance). However, as said, the corresponding predicted risks may still be biased in new patients, mainly due to different associations of the predictors in the new patients or to the presence of important predictors that were not included in the model. In case of the latter, adjustment of the model intercept is often sufficient and preferable, as such recalibration removes bias of the original intercept while precision of the regression coefficients is sustained [3,20]. In our example, a simple adjustment of the intercept of the prediction model also resulted in deletion of a substantial part of the bias of the predicted risks, without losing precision of the other regression coefficients.

A few methodological issues need to be addressed. First, there was a large difference in incidence of severe postoperative pain (defined as an NRS score  $\geq 6$  at least once in the first hour after surgery) between the derivation set and the updating and test sets (62%, 36%, and 35%, respectively). Because the distribution of the predictor values was very similar between the data sets, the difference in incidence should be explained by characteristics that were not included in the prediction model. One explanation could be the different time points at which the NRS pain scores were assessed. In the derivation set, pain was scored every 15 minutes after arrival at the PACU (15, 30, 45, 60 minutes). In the updating and test sets, pain of the inpatients was scored at 15 and 60 minutes, and of the outpatients at 15, 30, and 45 minutes after arrival at the PACU. The lower number of observations may have led to unobserved cases of severe postoperative pain, and therefore, to a lower observed incidence of severe postoperative pain. However, we estimated what the incidence of severe postoperative pain would be in the derivation set if the measurements were done on the same time points as in the updating set and test set. For the inpatients, the incidence of severe postoperative pain was 56% (compared to 60%), and for the outpatients 43% (compared to 44%). Hence, the lower number of pain measurements could only to a minor extent have led to the lower incidence of severe postoperative pain in the updating and test sets. A more plausible explanation is the difference or change in characteristics that were not included in the original model but do influence the originally estimated associations of the predictors in the model, for example, the change of pain management over time. It may well be that in the later period more aggressive preventative pain management was introduced in daily practice, caused by an increased focus on postoperative outcomes (due to new pain guidelines issued in 2003).

Second, the classification of type of surgery (five groups) was developed in the derivation study. Starting with 27 groups of surgical procedures (based on clinical experience, current practice, and interviews with surgeons and anesthesiologists), the univariable association between each surgical group and severe acute postoperative pain was estimated. Groups with similar associations were then combined, resulting in the five categories of surgery. The classification was thus partly data-driven and overfitting may have occurred [13]. To correct for this overfitting, the regression coefficients of type of surgery should have been extra shrunk. We therefore did an additional analysis, in which we shrunk the regression coefficients of the four indicator variables for type of surgery with the heuristic shrinkage factor of van Houwelingen that was estimated at 0.85 [19]. This shrinkage factor was calculated by taking into account the difference in  $\chi^2$  of the model with 27 surgical procedures and the model with the five categories of surgery, and the difference in degrees of freedom. However, when we applied the model with the extra shrunk regression coefficients of type of surgery, we found a similar performance as with the original prediction model (data not shown).

Third, it is often stated that the performance of models (developed or updated) should not be tested by split-sample methods, but rather by bootstrapping or several (randomly selected) cross validations [13,21,22] as chance findings are more likely to occur with the former. However, we explicitly used the split-sample method based on time to cohere with the natural chronology in practice. It is possible, however, that the large drop in AUC in the test set of the updated models could indeed be due to chance or unfortunate split sampling.

Finally, we left out two other possible updating approaches. The first is to rebuild a complete new model

starting from no prior knowledge and thus selecting the most important predictors based on the new data only. We explicitly did not apply this approach as there is ample evidence that this leads to overfitted prediction models with doubtful reproducibility in subsequent patient samples, certainly when the validation set is relatively small as often is the case [3,20]. The second approach is to retain the predictors of the original model and assess the added value of new predictors. Addition of new predictors may particularly increase the model's discriminative ability. For our example study, additional predictors of severe postoperative pain that are mentioned in the literature could be body mass index, type of anesthesia, and duration of surgery. However, we previously found that these had no predictive value in the derivation set [8,23]. Besides that, duration of surgery was only assessed postoperatively in the updating and test sets, and the preoperatively expected duration was not available. We note that in other situations it may be worthwhile to study the effect of adding predictors to an existing model, which can easily be done with method 3.

To conclude, in agreement with previous studies, our example study also demonstrates with empirical data that prediction models can be updated with simple recalibration methods. These simple recalibration methods improved the calibration to a similar extent as the revision methods that made more extensive adjustments to the original model. Discrimination was not improved by any of the methods. More generally speaking, we stress that before any model is applied in practice, we strongly advise that any model (diagnostic or prognostic) is first validated and, if necessary, updated. A validated (and, if necessary, updated) model may cautiously be applied in new patients who are similar to the patients in the derivation and validation sets. However, when the user has reasons to believe that the model may perform differently in the new patients,

## Appendix

Intercept and regression coefficients of the updated models

|                                            | Method 0 <sup>a</sup> | Method 1 | Method 2 | Method 3 | Method 4 | Method 5 |
|--------------------------------------------|-----------------------|----------|----------|----------|----------|----------|
| Intercept                                  | −0.42                 | −1.40    | −1.19    | −1.43    | −1.61    | −1.38    |
| Ambulatory surgery                         | −0.70                 | −0.70    | −0.53    | −0.63    | −0.88    | −0.79    |
| Female gender                              | −0.004                | −0.004   | −0.003   | −0.004   | 0.29     | 0.16     |
| Age                                        | −0.009                | −0.009   | −0.007   | −0.008   | −0.003   | −0.01    |
| Preoperative pain score                    | 0.11                  | 0.11     | 0.08     | 0.10     | 0.12     | 0.10     |
| Low pain surgery                           | 0.50                  | 0.50     | 0.38     | 0.45     | 0.59     | 0.70     |
| Moderate pain surgery                      | 0.92                  | 0.92     | 0.70     | 0.83     | 0.57     | 0.94     |
| High pain surgery                          | 1.05                  | 1.05     | 0.79     | 0.95     | 0.82     | 1.00     |
| Highest pain surgery                       | 1.72                  | 1.72     | 1.30     | 1.55     | 1.41     | 1.62     |
| Expected incision size ≥ 10 cm             | 0.39                  | 0.39     | 0.29     | 0.35     | 0.50     | 0.73     |
| APAIS anxiety score                        | 0.05                  | 0.05     | 0.04     | 0.05     | 0.01     | 0.03     |
| APAIS need for information                 | −0.05                 | −0.05    | −0.04    | −0.05    | −0.02    | −0.03    |
| Ambulatory surgery × female gender         | 0.67                  | 0.67     | 0.50     | 0.60     | 0.56     | 0.49     |
| Ambulatory surgery × low pain surgery      | −0.10                 | −0.10    | −0.08    | −0.57    | −0.44    | −0.17    |
| Ambulatory surgery × moderate pain surgery | −0.47                 | −0.47    | −0.35    | 0.41     | 0.70     | 0.04     |
| Ambulatory surgery × high pain surgery     | −0.07                 | −0.07    | −0.05    | 0.24     | 0.55     | 0.15     |
| Ambulatory surgery × highest pain surgery  | −1.51                 | −1.51    | −1.14    | 0.85     | 0.82     | −0.30    |

<sup>a</sup> Original prediction rule.

data of the new patients should first be collected to test the performance before the model can be applied in daily clinical practice. Any model may perform slightly different in every new patient sample due to sampling variation. In that situation, the model does not need to be updated. However, two questions remain: when is the difference in model performance large and is updating needed, and when has a model been sufficiently updated? So far, this particular methodological area of prediction research has not been explored. Future research should address the question how many validation studies and what type of adjustments are needed before it is justified to implement a prediction model into clinical practice.

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